

Siyuan Chen

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Research Interests

My research develops **AI foundation models for protein structure, interaction, and dynamics**, combining geometric deep learning, physics-grounded priors, and large-scale molecular dynamics data. I am particularly interested in (i) **physically valid biomolecular generation**, (ii) **scalable protein–protein interaction discovery**, and (iii) **Transformer-based neural simulators that learn protein conformational dynamics from MD trajectories**.

Education

University of British Columbia

Sep 2025 - Present

Ph.D. in Computer Science

Advisor: Prof. Peter Yichen Chen

Shanghai Jiao Tong University

Sep 2021 - Jun 2025

B.S. in Physics, Minor in Computer Science and Technology

GPA: 3.89 / 4.30 (Ranking 1 / 30; Average 90.31 / 100)

Exchange Student, University of Toronto (Sep - Dec 2024)

Research Experience

AI for Protein Structure, Interaction, and Dynamics

Jan 2025 - Present

- **Gauss–Seidel Projection for Biomolecular Complexes** (published at ICLR 2026): a differentiable projection that enforces physical validity (non-penetration, geometric consistency) on protein structure prediction foundation models, eliminating steric clashes in predicted biomolecular complexes while improving inference efficiency.
- **Protein-TetSphere: Registered Volumetric Geometry for Protein Representation Learning** (Oct 2025 - Present; preprint, arXiv:2609.36277): a representation-alignment pretraining framework that treats tetrahedral-sphere (TetSphere) protein-surface meshes as a third modality alongside sequence and atomistic structure, using surface geometry as a bridge to align sequence and structural embeddings for fast, scalable PPI discovery.
- **WorldParticle for Protein Dynamics Modeling** (SIGGRAPH Asia 2026): a unified Transformer-based simulator that treats proteins as Lagrangian particle systems and learns their dynamics directly from large-scale **molecular dynamics (MD) trajectory data**, enabling fast, generalizable neural surrogates for protein conformational dynamics across heterogeneous biomolecular systems.

Neural Surrogate Models for Physical Dynamics

2024 - Jan 2026

- **Factorized Neural-Implicit DMD for Parametric Dynamics** (preprint, 2026): a physics-coded neural-field operator that learns a Koopman / DMD spectral decomposition of parametric dynamical systems, factorizing spatial modes from temporal dynamics to enable interpretable, stable long-horizon prediction and zero-shot parameter generalization – methodology directly transferable to biomolecular and continuum dynamics.

- **Temporally-Aware Subspace Fluid Simulation** (published at SIGGRAPH 2025): a learned reduced-order surrogate for high-dimensional PDE dynamics; an online temporally-adaptive subspace captures emergent low-rank structure across time, yielding orders-of-magnitude speedups over full-order solvers while preserving long-horizon physical fidelity.

Selected AI4Science Publications & Preprints

- [1] **Siyuan Chen**, Cai Zhou, Jinrui Zhang, Zhaokang Liang, Taku Komura, Wojciech Matusik, Stephen Bates, Tommi Jaakkola, Wengong Jin, Peter Yichen Chen, and Minghao Guo. [From Surfaces to Volumes: Registered Geometry for Protein Representation Learning](#). *Preprint, arXiv:2609.36277*, 2026.
- [2] **Siyuan Chen**^{*}, Minghao Guo^{*†}, Caoliwen Wang, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen[†]. [Physically Valid Biomolecular Interaction Modeling with Gauss–Seidel Projection](#). *International Conference on Learning Representations (ICLR)*, 2026.
- [3] Caoliwen Wang^{*}, Minghao Guo^{*†}, **Siyuan Chen**^{*}, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen[†]. [WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer](#). *SIGGRAPH Asia (Conference Track)*, 2026.
- [4] **Siyuan Chen**^{*}, Zhecheng Wang^{*}, Yixin Chen, Yue Chang, Peter Yichen Chen, Eitan Grinspun, and Jonathan Panuelos[†]. [Factorized Neural Implicit DMD for Parametric Dynamics](#). *Preprint, arXiv:2603.10995*, 2026.
- [5] **Siyuan Chen**, Yixin Chen, Jonathan Panuelos, Otman Benckekroun, Yue Chang, Eitan Grinspun, and Zhecheng Wang[†]. [Fast Subspace Fluid Simulation with a Temporally-Aware Basis](#). *ACM Transactions on Graphics (SIGGRAPH)*, 2025.
- [6] Xingyu Ni, Jiong Chen[†], **Siyuan Chen**, Caoliwen Wang, Mathieu Desbrun, and Taku Komura. Efficient Open-Boundary Poisson Solves. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 2026.

Other Publications & Preprints

- [7] Shixun Huang^{*}, **Siyuan Chen**^{*†}, Yue Chang, Zhecheng Wang, and Peter Yichen Chen[†]. [Closing Trajectories: Equation-Free Cyclic Animation via Koopman Surrogates](#). *ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), Computer Graphics Forum*, 2026.
- [8] Yiwen Wang, **Siyuan Chen**, and Ran Yi[†]. [SG-Splatting: Accelerating 3D Gaussian Splatting with Spherical Gaussians](#). *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [9] F. Liu^{*}, Y. Takeuchi^{*}, Si Chen, **Siyuan Chen**, K. S. Khaw[†], M. Lyu, Z. Pan, D. Wang, et al. [Simulation Studies of a High-Repetition-Rate Electron-Driven Surface Muon Beamline at SHINE](#). *Physical Review Accelerators and Beams*, 28(8), 083401, 2025.
- [10] Y. Chen, T. Hu, Y. Tang, **Siyuan Chen**, A. Chen, and Ran Yi[†]. [Plasticine3D: 3D Non-Rigid Editing with Text Guidance by Multi-View Embedding Optimization](#). *arXiv preprint arXiv:2312.10111*, 2023.

Honors and Awards

Shanghai Jiao Tong University Third-Class Academic Scholarship (Top 15%), 2023

Third Prize, National College Student Mathematical Modeling Competition (Top 30%), 2022